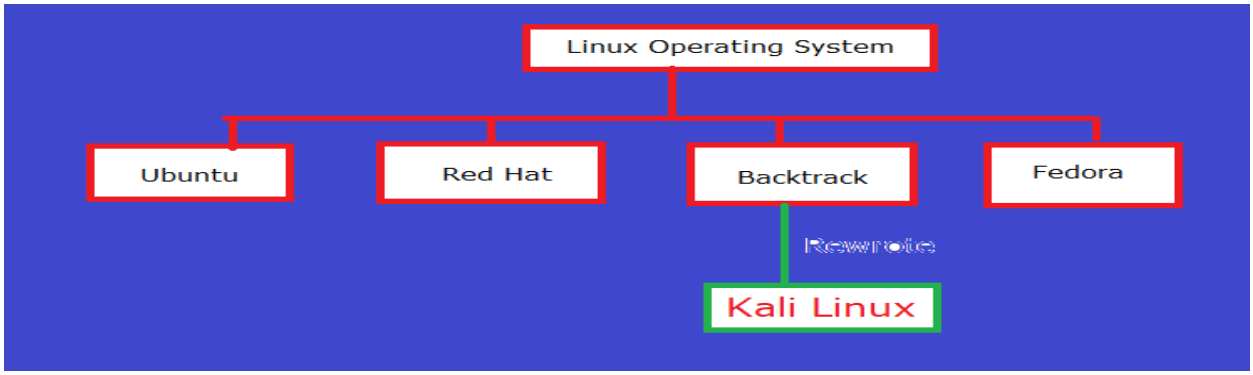
Kali Linux:

- ➔ Kali Linux is a powerful Linux distributed operating system designed for digital forensics and penetration testing. It is based on debian and developed by Offensive Security.
- ➔ Kali Linux provides vast number of predefined tools and a feature that makes task is very easy for cyber security professionals, ethical hackers and other security officers.
- ➔ Kali Linux was developed by Mati Aharoni and Devon Kearns of Offensive Security, who rewrote their previous distribution, BackTrack Linux, to create Kali.

Linux Distribution:

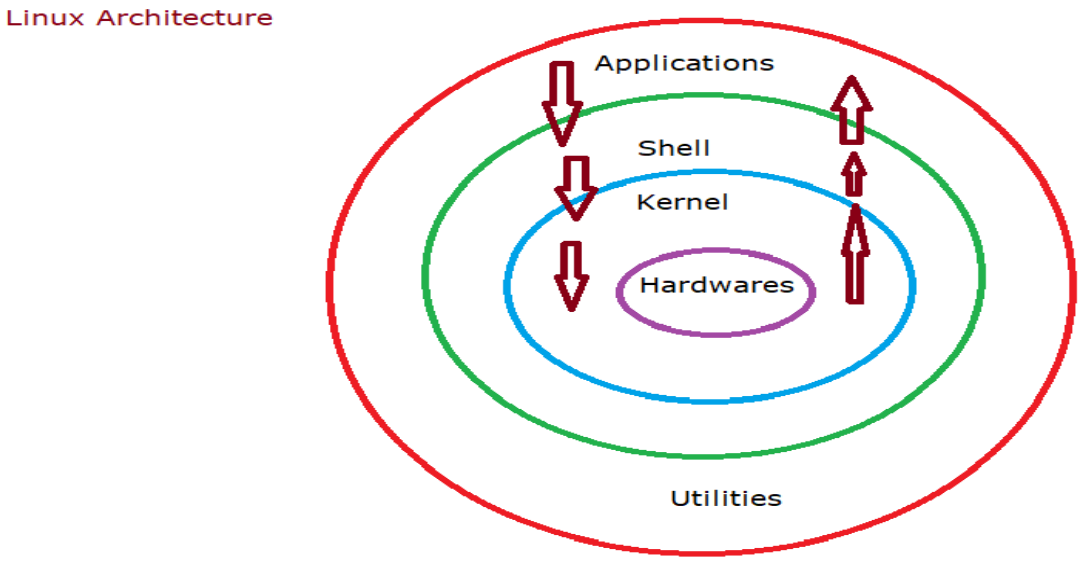
- ➔ In Linux distribution, we can have different Linux operating systems that are made up of kernel and adding special software's, utilities, packages that satisfied the user goals.



Operating System:

- ➔ Operating system is a software that act as interface between computer user applications (software's) and computer hardware's
- ➔ Operating system is a software that executes computer applications with the help of the hardware's
- ➔ Ex: Windows, macOS, Ubuntu, Kali, android etc….

Linux Architecture:



Shell:

- ➔ Shell is an interface that allows the user to interact with the operating system.
- ➔ They are two types on shells are available.
  1. Command Line Interface (CLI) Shell: This is text based interface where user interacts with the system by using commands.
     Ex: **B**ourne **A**gain **Sh**ell (BASH) commonly used in Linux. Windows Command Prompt and Power shell etc.

  2. Graphical Shell: This is visual based interface where user using icons and graphics to interacts with the system
     Ex: Windows Explorer

Kernel:

- ➔ Kernel is a core part of operating system that directly communicates with the hardware's like CPU, Memory, I/O devices etc.
- ➔ Kernel itself control and manage all the computer hardware's to run any application in the system.

Features of kali Linux:

- ➔ There are three major features of kali Linux, they are
  - ❖ Pre Installed Tools for Hacking
  - ❖ Open Source
  - ❖ More Security
  - ❖ Free of Cost
  - ❖ IP Tables (Firewall)

Virtual Machine (VMware):

- ➔ VMware is software that provides virtualization provides by which we can our computer with multiple operating systems.
- ➔ VMware use Hypervisor technology to provide virtualization in which all the computer hardware's like CPU, RAM or other devices become dual.
- ➔ Once we downloaded and installed VMware, in that we need to install kali Linux.

Kali Linux File System:

File systems arrange data in a hierarchical, tree-like structure, using directories (or folders) to group file.

- ➔ Some of the important directories are given below

  Bin Sbin Boot Dev etc lib media mnt opt proc root Run Ser

  Etc..
- ➔ Linux basic commands

  Ex: ls, cd, mkdir, nano, date, uptime, powerof, chmod etc…….